HARIPRASAD ANAND

haripanand04@gmail.com • +91 9995075462 • Alappuzha, Kerala, India (Open to Remote / Relocation) • LinkedIn • GitHub • Portfolio

PROFESSIONAL SUMMARY

Dynamic and results-driven Robotics and Automation Engineer with demonstrated expertise in ROS/ROS 2 autonomous navigation, ARM-based embedded firmware & PCB hardware bring-up, real-time computer vision pipelines, long-range RF & IoT telemetry. Experienced in the full product lifecycle from custom PCB design (KiCad) and ARM/STM32 board bring-up to autonomous navigation (ROS 2/Nav2/SLAM) and computer vision. Eager to contribute technical rigor and collaborative problem-solving to Asimov Robotics as a Robotics Software & Embedded Engineer.

TECHNICAL & PROFESSIONAL SKILLS

Embedded & Hardware:	STM32, ARM Cortex, ESP32, Arduino, Raspberry Pi, SPI / I2C / UART / USART, Modbus, RF modules (RFM69), PCB Design (KiCad), Hardware Debugging (ST-Link / SWD), PLC Programming
Robotics & Vision:	ROS / ROS 2, Nav2, SLAM, OpenCV, YOLO, Gazebo, UAV & UGV Development, Mission Planner, InsightFace
Software & Design:	Python, C / C++, TensorFlow, Flask, MQTT, CAD (SolidWorks), Linux / Bash, Tailscale, Git
Professional & Soft Skills:	Cross-functional Leadership, Team Management, Event Coordination, Communication, Technical Documentation

WORK EXPERIENCE

Robotics Engineer, R&D — SMEC Automation *(Kerala, India)* **July 2026 – Present**

- Design and bring up custom PCBs for ARM-based embedded products using KiCad, managing full lifecycle from schematic capture and layout through DRC review to fabrication.
- Execute hardware debugging and board bring-up utilizing ST-Link (SWD), serial consoles, and signal-level analysis (logic analyzer/oscilloscope) to isolate circuitry and firmware faults.
- Implement and debug mission-critical industrial and sensor communication protocols including SPI, USART/UART, I2C, and Modbus.
- Build, calibrate, and troubleshoot long-range RF communication links (915 MHz RFM69HCW) between transmitter and receiver boards with zero-packet-loss reliability.

KEY PROJECTS

Autonomous Mapping UGV — Lead Robotics & Embedded Developer | View Project

Tech: ROS 2, SLAM, Nav2, Raspberry Pi, STM32, UART

- Developed an autonomous ground vehicle utilizing ROS 2, Nav2, and SLAM for real-time 2D/3D environment mapping and autonomous navigation.
- Engineered reliable bidirectional UART motor control and sensor feedback telemetry between a Raspberry Pi companion computer and STM32 microcontroller.

AI Customer Care Chatbot & Vision Systems — AI / Software Developer | View Project

Tech: YOLO, OpenCV, LLMs, Groq, Ollama, Flask, Python

- Built an enterprise LLM support bot for SMEC Automation grounded in a proprietary knowledge base with human-escalation guardrails and a Flask web UI.
- Deployed computer vision systems including Plant Disease Detection (YOLO/OpenCV) and Target Identification & Warning System integrating ROS person re-identification with real-time alerting.

RF Warning System & Autonomous Agricultural Rover — Hardware & Firmware Engineer | View Project

Tech: STM32F103, RFM69HCW, GPS, SolidWorks, Sensors

- Designed a Coast Guard alarm link pairing STM32F103 MCUs over 915 MHz RFM69HCW transceivers with instant LED and haptic vibration alerts.
- Constructed a skid-steer autonomous rover equipped with a vision-based 4-DOF manipulator, industrial GPS tracking, and soil analytics sensors.

MQTT Multi-Device Control System — IoT Developer | View Project

Tech: MQTT, Mosquitto, Tailscale, ESP32, ADB, Web UI

- Engineered a secure multi-device orchestration system via Mosquitto broker over Tailscale mesh VPN, enabling low-latency control and screen mirroring across mobile and desktop devices.

EDUCATION

B.Tech in Robotics and Automation — Toc H Institute of Science and Technology *(Kerala, India)* **2022 – 2026**

CGPA: 7.37 | Core Coursework: Embedded Systems, Robotic Kinematics, Control Systems, Machine Vision

Higher Secondary (CBSE Class XII & X) — SDVEMHSS *(Kerala, India)* **2020 – 2022**

Class XII: 86.2% | Class X: 86.6%

CERTIFICATIONS

- **Internship on Industrial Automation** — Bosch Rexroth (2025)
- **Workshop on Embedded Systems** — Srishti Robotics (2024)
- **Workshop on UAV (Unmanned Aerial Vehicles)** — Fuselage Innovations (2024)